Vehicle Preventive Maintenance System (Python Project)

Overview:
Developed a Python-based system to digitally manage vehicle records and maintenance activities, replacing manual tracking with an organized and efficient solution.

Objective:
To simplify vehicle maintenance tracking and provide useful insights like cost analysis and service scheduling.

Key Features:
Add and manage vehicle details (ID, type, status, last service date)
Record maintenance data (service date, cost, parts used, technician, etc.)
Search vehicles based on type or current status

Generate summaries:
Total service cost per vehicle
Failure analysis
Service interval tracking

Technical Details:
Built using Python
Used pandas for data processing and analysis
Data stored in Excel (.xlsx) files
Implemented a menu-driven command-line interface

Outcome:
Created a simple and effective system that organizes maintenance data, reduces manual effort, and helps in better decision-making through basic data analysis.

Project: Sustainable Cool Roofing Solutions using Green Chemistry Principles

Developed an eco-friendly and cost-effective cool roofing solution to reduce indoor heat in residential buildings by applying green chemistry principles. The project addresses rising temperatures and excessive heat absorption caused by conventional roofing materials.

Objective:
To design a low-cost, sustainable, and bio-based roof coating that improves solar reflectance and reduces heat transfer.

Key Contributions:
Designed a bio-based cool roof coating using natural materials like lime, rice husk ash, clay, and eggshell powder
Improved solar reflectance to minimize heat absorption
Proposed a non-toxic, environmentally friendly alternative to synthetic coatings
Focused on affordability and easy application for residential use

Core Concepts Used:
Green Chemistry Principles
Sustainable Material Design
Thermal Heat Transfer Basics
Research & Comparative Analysis

Impact
Reduced indoor temperature by approximately 2–4°C
Lowered cooling energy consumption by 10–20%
Improved indoor comfort and energy efficiency
Promoted sustainable and eco-friendly construction practices

Skills Gained:
Research and analytical thinking
Problem-solving approach
Basic understanding of sustainable technologies
Team collaboration and communication
Presentation and technical documentation skills

Project: Solar-Powered EV Charging Parking System

Description:

Designed a sustainable and efficient parking system for electric two-wheelers by integrating solar-powered charging stations with shaded parking infrastructure. The project focuses on promoting green mobility and improving parking facilities within a campus environment.

Key Contributions:

Developed a prototype model of a solar-powered shaded parking system
Integrated renewable energy (solar panels) for EV charging
Designed an organized parking layout for better space utilization
Applied design thinking methodology (empathy, ideation, prototyping, testing)
Conducted user testing and collected feedback for improvements

Impact:

Promotes eco-friendly transportation and sustainability
Reduces dependency on conventional energy sources
Improves user convenience, parking efficiency, and vehicle protection
Supports reduction in carbon emissions

Skills Gained:

Problem-solving, Design Thinking, Research & Analysis, Team Collaboration, Basic Renewable Energy Concepts, Communication & Presentation Skills

AI Career Analyzer & Resume Intelligence Platform (Ongoing Project)

Project Overview:
The AI Career Analyzer & Resume Intelligence Platform is a web-based application that provides personalized career guidance and resume optimization using Artificial Intelligence. It helps students and job seekers identify suitable career paths and improve their resumes based on their skills and qualifications.

Objective:
To assist users in making better career decisions by analyzing resumes and providing skill-based recommendations and career roadmaps.

Key Features:
Resume upload and AI-based analysis (PDF/DOC)
Skill gap identification and improvement suggestions
Personalized career recommendations
Career roadmap generation
AI chatbot for career guidance
User dashboard for insights

Methodology:
The system uses user inputs (education, interests, job preferences) along with resume data. AI models analyze this information to generate structured feedback, career suggestions, and improvement strategies.

Data Used:
Primary: User resumes and inputs
Secondary: Datasets from Kaggle, UCI, IndiaAI
Types: Structured (job roles, salary), Unstructured (resume text), Semi-structured (resume sections)

Technologies & Tools:
OpenAI API for AI-based analysis
Lovable for web app development
Supabase for database, authentication, and storage

Implementation Status:
Currently under development with focus on AI integration and core system features.
Future Scope:
Mobile app development, advanced AI assistant, and integration with learning platforms.

Skills Gained:
AI basics and API integration
Web development concepts
Database management (Supabase)
Data handling (structured & unstructured)
Problem-solving and teamwork